

Tensors, Honestly

An index is not decoration. It is a promise about what happens when someone else picks different coordinates.

WHERE WE ARE

Chapter 2.3 wrote x^μ and $\eta_{\mu\nu}$ and $ds^2 = \eta_{\mu\nu}dx^\mu dx^\nu$ and got away with it, on the strength of the fact that the formulas came out right. This chapter pays for that. Everything in 2.3 was legitimate; none of it was *defined*. We now go back and supply the definitions, and in the process the index notation stops being a bookkeeping habit and becomes an idea.

Here is the destination, stated in advance so you know what we are aiming at:

A tensor is defined by how its components change when you change coordinates.

Not by a picture, not by "a thing with indices", not by "a matrix with more slots". By a transformation law, and nothing else. That definition sounds like a technicality and is in fact the whole point, because it is what makes the following true: *any equation whose two sides transform the same way is automatically true in every frame if it is true in one*. That single sentence is why the rest of this book is written in indices. Section 6 proves it; everything before Section 6 is the machinery needed to state it; everything after is what to do with it.

One more thing to hold onto. The definition we are about to give never mentions Lorentz transformations. It works for *any* smooth change of coordinates — rotations, polar coordinates, an accelerating frame, a coordinate patch on a curved manifold. That generality is not showing off. It is the reason Part III can be about geometry rather than about notation: Chapter 3.2 will reuse this chapter's definition word for word, with only one change, and that change is the whole of general relativity.

TOOLS YOU'LL NEED — Chapter 0.4 §4: change of basis, the fact that components transform with P^{-1} while basis vectors transform with P , and $A' = P^{-1}AP$. That section is the entire skeleton of this one. Chapter 0.6 §4: the gradient is really a one-form, g_{ij} , and $(\nabla f)^i = g^{ij}\partial_j f$ — this chapter is the promised payoff of that section, and we pick the thread up by name in §3. Also 0.6 §5, the multivariable chain rule, which does almost all of the work below. Chapter 0.7 §4.3, the unique split of a matrix into symmetric plus antisymmetric parts, and §6, the continuity equation. Chapter 2.2 for the boost and rapidity; Chapter 2.3 for the interval, $\eta_{\mu\nu}$, and the statement that a four-vector is something transforming like dx^μ — which is the statement we are about to turn into a definition.

1 · The problem, stated sharply

Physics is not allowed to depend on the coordinates a human being chose. Nothing about the electron knows whether you set up your axes pointing north or north-north-east, or whether you are drifting past the laboratory at $0.6c$. Whatever is real must be the same for everyone.

But *components* obviously are not the same for everyone. That is what components are: numbers you get by comparing a physical thing to a set of reference directions you invented. Chapter 0.4 §4 made this precise for an ordinary vector — change the basis by P and the components change by P^{-1} — and Chapter 2.2 made it precise for spacetime, where changing to a moving frame reshuffles the four numbers (ct, x, y, z) by a boost.

So we have a tension. The things we can write down are components, which are not invariant. The things we want to assert are laws, which must be. The resolution is not to abandon components. It is to insist on a particular class of objects:

THE WHOLE STRATEGY

Work only with objects whose components change in a **controlled, known way** — a way fixed entirely by the coordinate change and not at all by the object. Then build statements out of them in which the changes cancel. The pieces are frame-dependent; the statement is not.

That is a design constraint, and like all good design constraints it rules things out. Let us see it rule something out, because the failure is more instructive than the success.

1.1 · Four numbers that are not a four-vector

Here is an object that has an index and is not a tensor. Define

$$N^\mu \equiv (1, 0, 0, 0) \quad \text{in every inertial frame.} \quad (2.4.1)$$

Nothing in that definition is ill-formed. It is a perfectly good rule: hand me a frame, I hand you four numbers. You could imagine it standing for "the direction of time", or — historically, and this is exactly the mistake Chapter 2.1 dismantled — the rest frame of the ether. It has an upper index, it has four components, it looks like everything else we write. Watch what happens.

Take a particle with four-momentum $p^\mu = (E/c, p^x, p^y, p^z)$, a genuine four-vector. Build the number

$$\mathcal{E} \equiv \eta_{\mu\nu} N^\mu p^\nu = N^0 p^0 - \mathbf{N} \cdot \mathbf{p} = \frac{E}{c}. \quad (2.4.2)$$

It *looks* like a scalar: indices all matched and summed, nothing dangling. Now evaluate it in two frames using specific numbers. Let $p^\mu = (5, 3, 0, 0)$ in units of MeV/c , and let S' move at $\beta = 3/5$, so $\gamma = 5/4$. Chapter 2.2's boost gives $p'^\mu = (4, 0, 0, 0)$ — we will grind that arithmetic out in §9, but you can check it in your head: $\frac{5}{4}(5 - \frac{3}{5} \cdot 3) = \frac{5}{4} \cdot \frac{16}{5} = 4$. Then

$$\mathcal{E}|_S = 1 \cdot 5 = 5, \quad \mathcal{E}|_{S'} = 1 \cdot 4 = 4. \quad (2.4.3)$$

Two observers, one recipe, two answers. And note carefully what did *not* go wrong: no algebra was botched, no sign was dropped. The recipe $\eta_{\mu\nu} N^\mu p^\nu$ is unambiguous and both computations are correct. What failed is the claim that the recipe defines *a number*. It defines a number per frame, which is to say it defines nothing at all — you cannot write " $\mathcal{E} = 5$ " and mean anything by it.

The damage gets worse if you write an equation. Consider

$$p^\mu = mc N^\mu. \quad (2.4.4)$$

In frame S' , where our particle is at rest, that is true: $p'^\mu = (4, 0, 0, 0)$ and $mc = 4$. If N^μ were a four-vector, the theorem of §6 would let us conclude that (2.4.4) holds in *every* frame — and since N^μ has been declared to be $(1, 0, 0, 0)$ everywhere, that says $p^\mu = (mc, 0, 0, 0)$ everywhere: this particle is at rest for every observer, including the one watching it go past at $0.6c$. The conclusion is false, so one of the premises is. It is not the theorem. It is the assumption that N^μ is a four-vector, and the theorem's one hypothesis is exactly the one N^μ fails.

△ WHY THIS ISN'T OBVIOUS

The array $(1, 0, 0, 0)$ is a perfectly respectable set of components. There is a genuine four-vector with those components — several, in fact; the four-velocity of any observer, divided by c , in that observer's own frame. The error in (2.4.1) is not the numbers. **The error is the phrase "in every frame."**

Specify a four-vector by giving its components *in one frame* and you have said something. The transformation law then tells everyone else what they will measure, and nobody is free to disagree. Specify it by decreeing the components in all frames at once and you have over-determined the object: you have made four independent statements where the geometry only permitted one, and the extra three are false.

This is the single most common way that beginners' index expressions go wrong, and it survives into professional work in subtler disguises. Any time you find yourself saying "in this frame the components are simply ..." and then *using that form in another frame*, you have made this mistake.

So the slogan for the chapter, which you should read as a definition and not as a warning:

WHAT A TENSOR IS NOT

A tensor is not "a thing with indices". It is a thing with indices that transform correctly. The indices are not a filing system for numbers. They are a claim, and the claim is checkable.

2 · Vectors, defined by how they transform

We need a prototype: one object whose transformation law we can compute rather than choose. There is exactly one natural candidate, and Chapter 2.3 already leaned on it.

Let x^μ be one set of coordinates and x'^μ another, related by four functions $x'^\mu = x'^\mu(x^0, x^1, x^2, x^3)$ which are smooth and invertible. Take two nearby events separated by an infinitesimal displacement dx^μ . What is the separation in primed coordinates? This is not a matter of convention; it is the multivariable chain rule of Chapter 0.6 §5, applied to each of the four functions x'^μ :

$$dx'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} dx^\nu, \quad (2.4.5)$$

summed over $\nu = 0, 1, 2, 3$. Nothing was assumed. Both sides are the same physical displacement between the same two events, described twice. Give the array of partial derivatives a name:

$$\Lambda^\mu{}_\nu \equiv \frac{\partial x'^\mu}{\partial x^\nu}, \quad \text{so that} \quad dx'^\mu = \Lambda^\mu{}_\nu dx^\nu. \quad (2.4.6)$$

This is the Jacobian matrix of Chapter 0.6 §8, wearing index notation. The upper index labels the row and the new coordinate; the lower index labels the column and the old one. Keep the horizontal offset in $\Lambda^\mu{}_\nu$ — it says which slot comes first, and by §4 that will start to matter.

Now make the leap that organises everything after it. We have one object, dx^μ , that we know transforms by (2.4.6). We *define* the class of objects that behave the same way.

DEFINITION — CONTRAVARIANT VECTOR

A **contravariant vector** (or just *vector*, or in four dimensions a *four-vector*) is an object with four components V^μ per coordinate system, such that under any change of coordinates

$$V'^\mu = \Lambda^\mu{}_\nu V^\nu, \quad \Lambda^\mu{}_\nu = \frac{\partial x'^\mu}{\partial x^\nu}.$$

That is the entire definition. There is no additional requirement, and no picture is part of it.

Two things should feel unsatisfying about that, and both are worth confronting.

"It's circular — you defined vectors by copying dx ." Yes, deliberately. The definition is *ostensive*: it points at a known object and says "these ones". What makes it non-vacuous is that many other objects turn out to belong to the class, and their membership is a theorem rather than a stipulation. The four-velocity $u^\mu = dx^\mu/d\tau$ is one, because $d\tau$ is a scalar (Chapter 2.3) and dividing a vector by a scalar leaves the law (2.4.6) intact. The four-momentum $p^\mu = mu^\mu$ is one, since m is a scalar by construction — in this book m always means *rest* mass, the number every observer computes from $p_\mu p^\mu = m^2 c^2$. The current $J^\mu = (\rho c, \mathbf{J})$ is one, and §9 shows what that buys. Each of those is a claim about physics, checkable and falsifiable, and each of them fails for N^μ of (2.4.1).

"Where did the arrow go?" Nowhere — it is still available, and Chapter 0.4 tells you exactly how. There, a vector was an element of a vector space and its components were x_j in a basis $\{e_j\}$; changing basis by $e'_j = P_{ij}e_i$ sent the components to $\mathbf{x}' = P^{-1}\mathbf{x}$. Compare that with the definition above and read off the dictionary:

$$\Lambda = P^{-1}. \quad (2.4.7)$$

The components transform with Λ , the basis vectors with $\Lambda^{-1} = P$, and their product — the actual arrow — does not move at all. Chapter 0.4 §4 put it in one sentence: *make your ruler twice as long and every measurement in rulers is halved*. "Contra-variant" is literally that: the components vary contrary to the basis. So the arrow is exactly as real as it ever was. What has changed is which end of the relationship we take as the definition, and we take the components' end because it is the end that still makes sense when there is no global vector space to draw arrows in — which, from Chapter 3.2 onward, is always.

2.1 • What Λ is for a boost

Specialise to the transformation of Chapter 2.2. With S' moving along $+x$ at speed $v = \beta c$ and $x^\mu = (ct, x, y, z)$,

$$ct' = \gamma(ct - \beta x), \quad x' = \gamma(x - \beta ct), \quad y' = y, \quad z' = z. \quad (2.4.8)$$

Differentiate each new coordinate with respect to each old one. The coefficients are constants, so the partial derivatives are just those constants:

$$\Lambda^\mu{}_\nu = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \gamma = \frac{1}{\sqrt{1 - \beta^2}}. \quad (2.4.9)$$

That is the matrix Chapter 2.3 wrote down. It is now not a postulate but a computed Jacobian, and its constancy — every entry independent of position — is a special feature of Lorentz transformations, not a general one. Hold onto that observation; it is the hinge on which §9's second worked example turns and it is the single technical difference between this chapter and Chapter 3.3.

▼ Grind box — the inverse boost, and why rapidity is the sane variable

Invert (2.4.8) by solving for ct and x . Multiply the first by γ and the second by $\gamma\beta$ and add:

$$\gamma ct' + \gamma\beta x' = \gamma^2(1 - \beta^2) ct = ct,$$

using $\gamma^2(1 - \beta^2) = 1$, which is the definition of γ rearranged. The same manoeuvre gives $x = \gamma(x' + \beta ct')$. So

$$(\Lambda^{-1})^\mu{}_\nu = \frac{\partial x^\mu}{\partial x'^\nu} = \begin{pmatrix} \gamma & +\gamma\beta & 0 & 0 \\ +\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \Lambda(-\beta),$$

which is the physically obvious statement that if S' moves at $+v$ relative to S then S moves at $-v$ relative to S' — now derived rather than assumed.

Chapter 2.2 introduced the **rapidity** ϕ with $\tanh \phi = \beta$, whence $\gamma = \cosh \phi$ and $\gamma\beta = \sinh \phi$ (divide $\cosh^2 - \sinh^2 = 1$ by $\cosh^2$ to see the first). In that variable

$$\Lambda(\phi) = \begin{pmatrix} \cosh \phi & -\sinh \phi & 0 & 0 \\ -\sinh \phi & \cosh \phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

and the hyperbolic addition formulas give, in two lines of matrix multiplication,

$$\Lambda(\phi_2) \Lambda(\phi_1) = \Lambda(\phi_1 + \phi_2).$$

Rapidities add. That is why ϕ is the right variable and β is not, and it is also a consistency check on the whole framework: the chain rule guarantees that composing coordinate changes composes their Jacobians,

$$\frac{\partial x''^\mu}{\partial x^\nu} = \frac{\partial x''^\mu}{\partial x'^\rho} \frac{\partial x'^\rho}{\partial x^\nu},$$

so the transformation law of §2 is automatically consistent under composition. Nothing extra had to be imposed. Chapter 6.1 will call this "the transformations form a group and the components carry a representation of it"; here it is just the chain rule.

3 · Covectors — the other kind of index

Chapter 0.6 §4 ended with a promise: *differentiation naturally produces lower indices, vectors naturally carry upper ones, the metric converts between them, and when the metric is the identity you cannot see the conversion happening*. It said that Chapter 2.4 would make this a formal convention. Here we are.

Take a scalar field f — a function on spacetime whose value at an event is the same number for everyone, $f'(x') = f(x)$. Temperature at a point, say, or the proper time elapsed along some fixed worldline. Its four partial derivatives

$$\partial_\mu f \equiv \frac{\partial f}{\partial x^\mu} \tag{2.4.10}$$

are perfectly good components of something. Which something? Compute. Apply the chain rule to f regarded as a function of the primed coordinates:

$$\partial'_\mu f = \frac{\partial f}{\partial x'^\mu} = \frac{\partial x^\nu}{\partial x'^\mu} \frac{\partial f}{\partial x^\nu} = \frac{\partial x^\nu}{\partial x'^\mu} \partial_\nu f. \quad (2.4.11)$$

Look at what appeared. Not $\frac{\partial x'^\mu}{\partial x^\nu}$, but $\frac{\partial x^\nu}{\partial x'^\mu}$ — the derivatives of the *old* coordinates with respect to the *new*. That is the inverse Jacobian, and we can prove it is the matrix inverse of Λ without computing anything, again by the chain rule:

$$\frac{\partial x'^\mu}{\partial x^\nu} \frac{\partial x^\nu}{\partial x'^\rho} = \frac{\partial x'^\mu}{\partial x'^\rho} = \delta^\mu_\rho, \quad (2.4.12)$$

since the primed coordinates are independent of one another: $\partial x'^0 / \partial x'^0 = 1$ and $\partial x'^0 / \partial x'^1 = 0$, and so on. So writing $(\Lambda^{-1})^\nu_\mu \equiv \frac{\partial x^\nu}{\partial x'^\mu}$, the notation is honest — that array really is the inverse matrix of (2.4.6), and (2.4.12) reads $\Lambda \Lambda^{-1} = I$.

So $\partial_\mu f$ transforms with the inverse of Λ , while dx^μ transforms with Λ . These are different laws. They give different numbers. Objects obeying the second one deserve their own name and their own index position:

DEFINITION — COVARIANT VECTOR (ONE-FORM, COVECTOR)

A **covector** is an object with four components ω_μ per coordinate system such that under any change of coordinates

$$\omega'_\mu = (\Lambda^{-1})^\nu_\mu \omega_\nu, \quad (\Lambda^{-1})^\nu_\mu = \frac{\partial x^\nu}{\partial x'^\mu}.$$

We write its index **down**. The gradient $\partial_\mu f$ of any scalar is the prototype, exactly as dx^μ was the prototype for vectors.

3.1 · Why the two species exist: the pairing

Now the structural fact that justifies having invented a second kind of object. Take a vector V^μ and a covector ω_μ and contract them — sum over the repeated index, one up and one down:

$$\begin{aligned} V'^\mu \omega'_\mu &= \left(\Lambda^\mu_\nu V^\nu \right) \left((\Lambda^{-1})^\rho_\mu \omega_\rho \right) \\ &= \underbrace{\Lambda^\mu_\nu (\Lambda^{-1})^\rho_\mu}_{= \delta^\rho_\nu} V^\nu \omega_\rho = \delta^\rho_\nu V^\nu \omega_\rho = V^\nu \omega_\nu. \end{aligned} \quad (2.4.13)$$

The two Jacobians met and collapsed to a Kronecker delta, and the delta then just renamed an index. So $V^\mu \omega_\mu$ takes the **same value in every coordinate system**. It is a scalar — a real number, not a number-per-frame — and it is the first genuinely invariant thing we have built, with no metric required.

Read the two definitions again in the light of that calculation. The vector law and the covector law are not two arbitrary conventions that happen to be inverse. Covectors are *defined* to be the objects that pair with vectors to give invariants; that is what they are for, and the inverse Jacobian is forced on them by that job. In Chapter 0.4's language, the covectors at a point form the **dual space** V^* : the space of linear maps from vectors to numbers.

3.2 · The picture: arrows and stacks

There is a genuine geometric image here, and it is not the one you would guess.

A vector is an arrow. Fine. A covector is *not* an arrow. A covector ω is a machine that eats an arrow and returns a number, linearly; and the honest picture of such a machine is a family of **evenly spaced parallel surfaces** — the level sets $\omega = 0, 1, 2, 3, \dots$ of the linear function it computes. In two dimensions they are parallel lines; in four they are parallel hyperplanes. A "big" covector has closely spaced surfaces, a "small" one has widely spaced surfaces; the covector's magnitude lives in the *density* of the stack, which is why doubling ω halves the spacing.

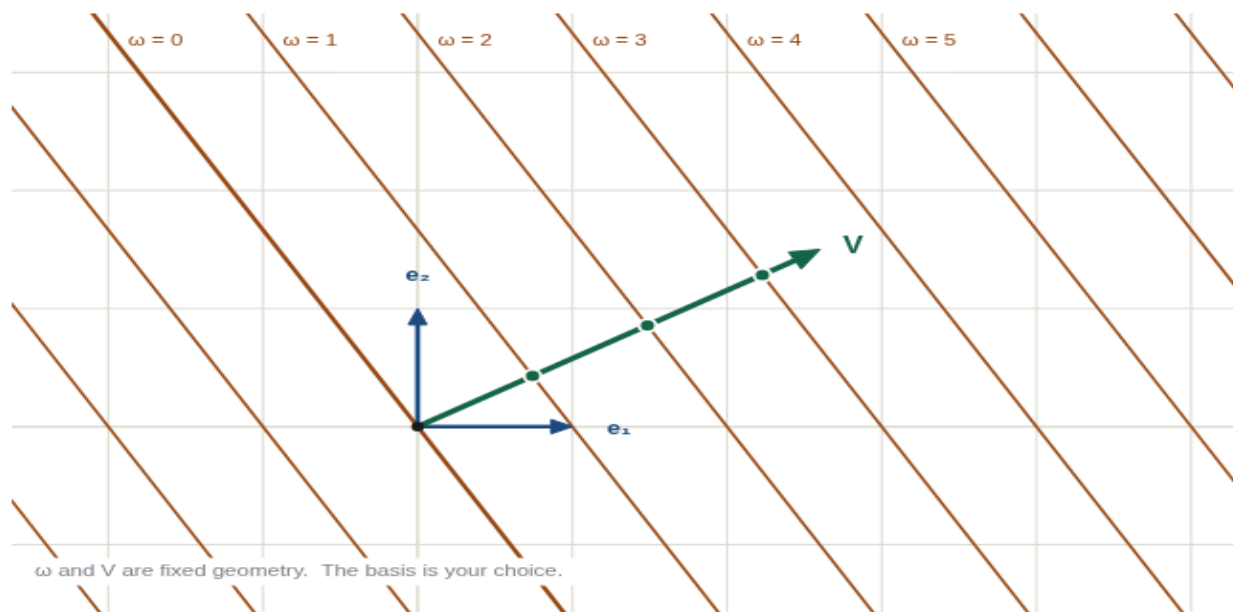
And then the contraction $V^\mu \omega_\mu$ has a completely concrete meaning:

WHAT CONTRACTION COUNTS

$V^\mu \omega_\mu$ is **the number of ω -surfaces the arrow V pierces**.

An arrow either crosses a given surface or it does not. No coordinate system has an opinion about that. This is why the pairing is invariant, in one sentence and with no algebra.

For the gradient this picture is one you have already seen without knowing it: the level sets of df are the contours of f , and $V^\mu \partial_\mu f$ is the number of contour lines you cross walking along V — which is, of course, the change in f . Chapter 0.6's contour figure was a picture of a one-form all along.



skew $\angle(\mathbf{e}_1, \mathbf{e}_2)$ 90.0° $|\mathbf{e}_1|$ 1.00

$|\mathbf{e}_2|$ 1.00

show the wrong pairing

reset to Cartesian

$V^1 = 2.60000$ $V^2 = 1.50000$ $\omega_1 = 1.00000$ $\omega_2 = 0.60000$

$V^i \omega_i = 3.50000000$ (drift 0.0e+0)

lines pierced = 3 (tip sits at $\omega = 3.50$)

Two species, one number. The green arrow is a vector V ; the orange stack is a covector ω , drawn as its level lines $\omega = 0, 1, 2, \dots$. Both are fixed geometric objects and neither moves when you touch a slider. There is deliberately no background grid to anchor them to — the only coordinate system on the page is the blue one, $\{\mathbf{e}_1, \mathbf{e}_2\}$ and its lattice, and that one is your arbitrary choice. The readouts are the components of V and of ω in that basis, and their contraction. **Start with $|\mathbf{e}_1|$.** Double it: V^1 halves, from 2.6 to 1.3, while ω_1 doubles, from 1 to 2. That is Λ and Λ^{-1} , visible in two numbers — measure in longer units and a vector's components shrink while a covector's grow, because a longer step crosses more of ω 's lines. Their product is untouched. Now skew the basis and rescale $\mathbf{e}_2$ as well: all four components wander, some of them through zero and into negative values, and $V^i \omega_i$ sits at 3.50000000 throughout, with a residual of at most a few parts in 10^{16} that is floating-point roundoff and nothing else. That is (2.4.13), measured rather than asserted. Meanwhile the ringed dots mark where the arrow crosses a level line: there are always exactly three, because the tip sits at $\omega = 3.5$, and no re-coordinatising can change how many lines a fixed arrow crosses. **Now press "show the wrong pairing".** That readout treats ω as though it were a vector — takes its *contravariant* components $\tilde{\omega}^i$ in the same basis and forms the naive sum $\sum_i V^i \tilde{\omega}^i$, in which both factors transform the same way. At the Cartesian start it agrees exactly, 3.500000, which is precisely why nobody notices the distinction in a first course. Then move a single slider and it goes: 1.550000 when you double $|\mathbf{e}_1|$, 2.333308 at 60° , and — with both basis vectors at their shortest and the angle at its widest — 77.144991. The up/down distinction is not decoration. It is the difference between a number and a number-per-frame.

3.3 · Why nobody told you this before

Because in Euclidean space with Cartesian coordinates the two species have *numerically identical components*, and so the distinction is invisible. Chapter 0.6 §4 showed why: converting between them requires the metric, and in Cartesian coordinates on flat Euclidean space $g_{ij} = \delta_{ij}$, so the conversion multiplies everything by 1. Every dot product you have ever computed as $\sum_i a_i b_i$ was silently a contraction of a vector with a covector, and you were spared the bookkeeping because the metric was doing nothing.

The invisibility ends the moment either of two things happens, and in relativity both do. Use non-Cartesian coordinates and g_{ij} stops being δ_{ij} (Chapter 0.6 §4 derived the polar gradient's notorious $1/r$ from exactly this). Or use a metric that is not positive-definite — $\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$ — and the conversion starts flipping signs. That is §4.

△ TWO DIFFERENT SPECIES OF OBJECT

V^μ and ω_μ are not two notations for the same thing. They live in different vector spaces, obey different transformation laws, and have different geometric pictures. An expression like $V^\mu + \omega_\mu$ is not merely bad style; it is meaningless, the way "3 metres + 3 kilograms" is meaningless. The sum would be one thing in one frame and something else in another.

The reason this is worth being pedantic about now is that in Part III the metric acquires position dependence and there is no coordinate system anywhere that makes the two coincide. Any habit of thought that relies on "they're the same really" fails there, permanently and silently.

4 · The metric as the dictionary

So far, no metric. Vectors and covectors and their pairing all exist without one. Now we add the object that Chapter 2.3 built from the invariance of the interval, and see exactly what it buys.

Chapter 2.3's central claim was that the quantity

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu, \quad \eta_{\mu\nu} = \text{diag}(1, -1, -1, -1), \quad (2.4.14)$$

is the same for all inertial observers. Write that out as an equation between two computations of the same thing, using (2.4.6) on the primed side:

$$\eta_{\mu\nu} dx'^\mu dx'^\nu = \eta_{\mu\nu} \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma dx^\rho dx^\sigma \stackrel{!}{=} \eta_{\rho\sigma} dx^\rho dx^\sigma. \quad (2.4.15)$$

Since dx^ρ is arbitrary, the coefficients must agree (symmetrised, which η already is), giving the defining property of a Lorentz transformation:

$$\boxed{\eta_{\mu\nu} \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma = \eta_{\rho\sigma}.} \quad (2.4.16)$$

In matrix form $\Lambda^T \eta \Lambda = \eta$. This is worth pausing on: it says the Lorentz transformations are exactly the linear maps that *preserve* η , in the same way that rotations are exactly the linear maps preserving δ_{ij} , i.e. $R^T R = I$. Chapter 2.3 derived the boost from the two postulates; (2.4.16) is that result restated as an algebraic condition, and it is the form Chapter 6.1 will generalise when it names the group $SO(1, 3)$.

4.1 · Lowering

Define, for any vector V^μ ,

$$V_\mu \equiv \eta_{\mu\nu} V^\nu. \quad (2.4.17)$$

The name is "lowering the index", and the claim implicit in writing the result with a down index is that V_μ is a genuine covector. That is a claim, so prove it. Start from the definition in the primed frame and push:

$$V'_\mu = \eta_{\mu\nu} V'^\nu = \eta_{\mu\nu} \Lambda^\nu_\rho V^\rho. \quad (2.4.18)$$

We want that to equal $(\Lambda^{-1})^\sigma_\mu V_\sigma = (\Lambda^{-1})^\sigma_\mu \eta_{\sigma\rho} V^\rho$. So the whole question is whether

$$\eta_{\mu\nu} \Lambda^\nu_\rho \stackrel{?}{=} (\Lambda^{-1})^\sigma_\mu \eta_{\sigma\rho}. \quad (2.4.19)$$

Take (2.4.16) and contract it with $(\Lambda^{-1})^\sigma_\alpha$ on the index σ :

$$\begin{aligned} \eta_{\rho\sigma} (\Lambda^{-1})^\sigma_\alpha &= \eta_{\mu\nu} \Lambda^\mu_\rho \Lambda^\nu_\sigma (\Lambda^{-1})^\sigma_\alpha \\ &= \eta_{\mu\nu} \Lambda^\mu_\rho \delta^\nu_\alpha = \eta_{\mu\alpha} \Lambda^\mu_\rho, \end{aligned} \quad (2.4.20)$$

which is (2.4.19) with the free indices renamed and the symmetry $\eta_{\mu\nu} = \eta_{\nu\mu}$ used on both sides. So yes: V_μ is a covector, and (2.4.17) is a legitimate map from one species to the other. Note precisely what made it work — (2.4.16), the invariance of η . Lowering with an arbitrary array of numbers would not have produced a covector.

4.2 · Raising

Going the other way needs the inverse. Since η is a non-degenerate matrix it has one; call its entries $\eta^{\mu\nu}$, *defined* by

$$\eta^{\mu\rho} \eta_{\rho\nu} = \delta^\mu_\nu. \quad (2.4.21)$$

For our signature the inverse is easy, because $\text{diag}(1, -1, -1, -1)$ squares to the identity: $\eta^{\mu\nu} = \text{diag}(1, -1, -1, -1)$ as an array of numbers, the same four numbers down the diagonal. That coincidence is a property of this particular metric and *not* a general fact — in Chapter 3.3 the inverse metric $g^{\mu\nu}$ is a genuinely different array, and confusing the two is a classic and expensive error. Raise with it:

$$V^\mu = \eta^{\mu\nu} V_\nu. \quad (2.4.22)$$

Consistent, since $\eta^{\mu\nu}V_\nu = \eta^{\mu\nu}\eta_{\nu\rho}V^\rho = \delta^\mu_\rho V^\rho = V^\mu$; raising after lowering returns you to where you started, which is the least one can ask of a dictionary.

4.3 · What it actually does to the numbers

Write (2.4.17) out. Since η is diagonal, only one term survives in each sum:

$$\begin{aligned} V_0 &= \eta_{00}V^0 = +V^0, & V_1 &= \eta_{11}V^1 = -V^1, \\ V_2 &= -V^2, & V_3 &= -V^3. \end{aligned} \tag{2.4.23}$$

So with signature $(+, -, -, -)$: **lowering flips the sign of the three spatial components and leaves the time component alone.** Concretely, for the four-momentum of \$1,

$$p^\mu = (5, 3, 0, 0) \quad \Longrightarrow \quad p_\mu = (5, -3, 0, 0), \tag{2.4.24}$$

and the invariant length is the contraction of the two:

$$p_\mu p^\mu = (5)(5) + (-3)(3) + 0 + 0 = 25 - 9 = 16 = (mc)^2, \tag{2.4.25}$$

so $m = 4 \text{ MeV}/c^2$. Note that $p_\mu p^\mu$ is a contraction of a covector with a vector, so by (2.4.13) it is invariant with no further argument needed — and §9 checks that by brute force in three different frames.

⚠ RAISING AN INDEX IS NOT MOVING A LETTER

The notation is so light that it invites you to think V^μ and V_μ are the same object written two ways, and that shifting the index up or down is a typographical act. It is not. **It is the application of a specific linear map** — the matrix η — and that map is not the identity. With our signature it multiplies three of the four components by -1 .

This is the single most productive source of sign errors in relativity, and the errors are the worst kind: dimensionally consistent, index-consistent, and off by a sign. Three habits inoculate you.

- **Never write an expression with two indices in the same position summed.** $V^\mu W^\mu$ is not a thing. It is not " $V \cdot W$ in a hurry"; it is a frame-dependent number, as §9 measures. If you want $V \cdot W$, write $\eta_{\mu\nu}V^\mu W^\nu$ or $V_\mu W^\mu$, which are the same thing.
- **Track where the minus signs are.** $V_\mu W^\mu = V^0 W^0 - \mathbf{V} \cdot \mathbf{W}$. The three-vector dot product enters with a minus. Every time.
- **∂^μ is not ∂_μ with the index moved.** It is $\eta^{\mu\nu}\partial_\nu$, so $\partial^0 = \partial_0$ but $\partial^i = -\partial_i$. Chapter 2.6's $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$ has signs in it that come from precisely here.

▼ Grind box — index gymnastics, and a bonus identity

Three working rules, each a one-line consequence of (2.4.21).

(i) You may slide a contracted pair up and down together.

$$V_\mu W^\mu = \eta_{\mu\nu} V^\nu W^\mu = V^\nu (\eta_{\nu\mu} W^\mu) = V^\nu W_\nu.$$

So $V_\mu W^\mu = V^\mu W_\mu$: it does not matter which factor carries the lowered index, only that exactly one does.

(ii) η with one index up and one down is the Kronecker delta. By (2.4.21), $\eta^\mu{}_\nu = \eta^{\mu\rho} \eta_{\rho\nu} = \delta^\mu{}_\nu$. So you never need to write $\eta^\mu{}_\nu$; it is δ .

(iii) The metric knows the inverse transformation. Define $\Lambda_\mu{}^\nu$ the only way the rules allow — lower the first index, raise the second: $\Lambda_\mu{}^\nu \equiv \eta_{\mu\alpha} \eta^{\nu\beta} \Lambda^\alpha{}_\beta$. Now contract the identity (2.4.20), namely $\eta_{\mu\alpha} \Lambda^\mu{}_\rho = \eta_{\rho\sigma} (\Lambda^{-1})^\sigma{}_\alpha$, with $\eta^{\rho\tau}$:

$$\begin{aligned} \text{left} &= \eta^{\rho\tau} \eta_{\mu\alpha} \Lambda^\mu{}_\rho = \Lambda_\alpha{}^\tau, \\ \text{right} &= \eta^{\rho\tau} \eta_{\rho\sigma} (\Lambda^{-1})^\sigma{}_\alpha = \delta^\tau{}_\sigma (\Lambda^{-1})^\sigma{}_\alpha = (\Lambda^{-1})^\tau{}_\alpha. \end{aligned}$$

Hence $\Lambda_\alpha{}^\tau = (\Lambda^{-1})^\tau{}_\alpha$. **Lowering and raising the indices of Λ produces its inverse.** This is why most books never write Λ^{-1} at all: they write $\Lambda_\mu{}^\nu$ and let the index positions carry the information. We have kept Λ^{-1} explicit up to this point because the covector law of §3 is true with no metric in sight, and it would have been dishonest to define it using one.

(iv) $\det \Lambda = \pm 1$. Take determinants of $\Lambda^\top \eta \Lambda = \eta$. By Chapter 0.4 §5, $\det$ is multiplicative and $\det \Lambda^\top = \det \Lambda$, so $(\det \Lambda)^2 \det \eta = \det \eta$, and $\det \eta = -1 \neq 0$ gives $(\det \Lambda)^2 = 1$. Boosts and rotations have $\det = +1$; a spatial reflection has $\det = -1$. That sign will matter exactly once in this chapter, in §8.

Two last remarks on η , both pointing forward.

It is what makes "length" mean anything. Without a metric there is no way to ask how long a four-vector is: you can pair a vector with a covector, but not a vector with a vector. η supplies exactly the missing ingredient, and the price is that "length squared" can be positive, negative, or zero — the timelike, spacelike, and null trichotomy of Chapter 2.3. The metric does not merely measure; it is what *defines* the causal structure.

It is about to become a field. Everything in this section used only that η is a symmetric, invertible, position-independent array satisfying (2.4.16). Drop the words "position-independent" — replace $\eta_{\mu\nu}$ by $g_{\mu\nu}(x)$, ten functions of position — and every formula above survives unchanged, while Λ becomes position-dependent and one thing breaks. What breaks is the derivative, as §9's second worked example shows. Repairing it is Chapter 3.3, and the repair is the gravitational field. *That is the entire difference between special and general relativity*, and it is a difference in one adjective.

5 · Tensors in general

Vectors carry one upper index and one factor of Λ . Covectors carry one lower index and one factor of Λ^{-1} . The generalisation writes itself, and it is a definition, not a theorem.

DEFINITION — TENSOR OF TYPE (P, Q)

A **tensor of type (p, q)** has p upper and q lower indices, hence 4^{p+q} components per coordinate system, and transforms with one factor of Λ per upper index and one factor of Λ^{-1} per lower index:

$$T'^{\mu_1 \dots \mu_p}_{\nu_1 \dots \nu_q} = \Lambda^{\mu_1}_{\alpha_1} \dots \Lambda^{\mu_p}_{\alpha_p} \times (\Lambda^{-1})^{\beta_1}_{\nu_1} \dots (\Lambda^{-1})^{\beta_q}_{\nu_q} T^{\alpha_1 \dots \alpha_p}_{\beta_1 \dots \beta_q}.$$

The law is *linear and homogeneous* in the components of T . Remember that; §6 is nothing but that observation.

The zoo so far, classified:

TYPE	OBJECT	EXAMPLE
$(0, 0)$	scalar	$m, d\tau, ds^2, p_\mu p^\mu$
$(1, 0)$	vector	$dx^\mu, u^\mu, p^\mu, J^\mu$
$(0, 1)$	covector	$\partial_\mu f, p_\mu, A_\mu$
$(0, 2)$	rank-2, both down	$\eta_{\mu\nu}, g_{\mu\nu}$ (Ch 3.3)
$(1, 1)$	mixed	$\delta^\mu_\nu, \partial_\nu V^\mu$
$(2, 0)$	rank-2, both up	$F^{\mu\nu}$ (Ch 2.6), $T^{\mu\nu}$ (Ch 3.6)

A scalar has no indices and therefore no factors of Λ : $\phi'(x') = \phi(x)$, the same number for everyone. That is consistent with the general law at $p = q = 0$, and it is why "scalar" and "invariant" are the same word here.

5.1 · The three operations that preserve tensor character

Tensors are useful because you can compute with them without leaving the class. Exactly three basic moves are permitted, and it is worth knowing that the list is short.

(1) Linear combination, same type. If S and T are both type (p, q) and a, b are scalars, then $aS + bT$ is type (p, q) . Immediate, because the transformation law is linear: apply it to each term and factor out the common string of Λ s. Note the restriction — you cannot add a $(1, 0)$ to a $(0, 1)$, for the reason given in §3.3.

(2) Outer product. If A^μ is type $(1, 0)$ and B_ν is type $(0, 1)$, then the sixteen numbers $C^\mu{}_\nu \equiv A^\mu B_\nu$ form a type $(1, 1)$ tensor. Proof: multiply the two transformation laws together and read off. In general, multiplying a (p, q) by an (r, s) component-by-component with all indices distinct gives a $(p + r, q + s)$.

(3) Contraction. Set one upper index equal to one lower index and sum. This is the interesting one, so we prove it.

THEOREM — CONTRACTION

If $T^\mu{}_\nu$ is a tensor of type $(1, 1)$, then $T^\mu{}_\mu$ (summed) is a scalar. More generally, contracting any upper index of a type- (p, q) tensor against any lower index yields a tensor of type $(p - 1, q - 1)$.

Proof of the basic case. Transform, then set the indices equal:

$$\begin{aligned} T'^\mu{}_\mu &= \Lambda^\mu{}_\alpha (\Lambda^{-1})^\beta{}_\mu T^\alpha{}_\beta \\ &= \delta^\beta{}_\alpha T^\alpha{}_\beta = T^\alpha{}_\alpha. \quad \blacksquare \end{aligned} \tag{2.4.26}$$

It is the same collapse as (2.4.13), and it is the same collapse every time: **a Λ and a Λ^{-1} sharing a summed index annihilate into a δ** . Once you have seen that once you have seen all of tensor algebra; the rest is bookkeeping about which indices are spectators.

▼ Grind box — contraction in the general case, with spectators

Take a $(2, 2)$ tensor $T^{\mu\nu}{}_{\rho\sigma}$ and contract ν against σ . Define $U^\mu{}_\rho \equiv T^{\mu\lambda}{}_{\rho\lambda}$ (sum on λ). Claim: U is a $(1, 1)$ tensor. Write the transformation law and set $\nu = \sigma$, calling the common value λ :

$$\begin{aligned} U'^\mu{}_\rho &= T'^{\mu\lambda}{}_{\rho\lambda} \\ &= \Lambda^\mu{}_\alpha \Lambda^\lambda{}_\beta (\Lambda^{-1})^\gamma{}_\rho (\Lambda^{-1})^\delta{}_\lambda T^{\alpha\beta}{}_{\gamma\delta}. \end{aligned}$$

Now the only factors carrying λ are $\Lambda^\lambda{}_\beta$ and $(\Lambda^{-1})^\delta{}_\lambda$, and λ is summed. Group them:

$$\begin{aligned} \Lambda^\lambda{}_\beta (\Lambda^{-1})^\delta{}_\lambda &= \delta^\delta{}_\beta, \\ U'^\mu{}_\rho &= \Lambda^\mu{}_\alpha (\Lambda^{-1})^\gamma{}_\rho \delta^\delta{}_\beta T^{\alpha\beta}{}_{\gamma\delta} = \Lambda^\mu{}_\alpha (\Lambda^{-1})^\gamma{}_\rho U^\alpha{}_\gamma, \end{aligned}$$

which is exactly the $(1, 1)$ law. The spectator indices μ and ρ never participated; they carried their own Λ and Λ^{-1} through untouched. The argument is identical for any (p, q) and any pair of indices, which is why nobody writes it out twice.

Two consequences worth stating. Contracting a vector with a covector is the case $(1, 0) \otimes (0, 1) \rightarrow (0, 0)$: that is (2.4.13) again. And $p_\mu p^\mu = \eta_{\mu\nu} p^\mu p^\nu$ is $(0, 2) \otimes (1, 0) \otimes (1, 0)$ contracted twice down to

$(0, 0)$, which is why it is a scalar — not because someone declared mass invariant, but because of the index structure.

△ WHY THE SUMMATION CONVENTION ONLY EVER PAIRS UP WITH DOWN

Because that is the only pairing whose result is a tensor. Λ against Λ^{-1} collapses to δ ; Λ against Λ does not collapse to anything, and what you get depends on the frame. So an expression like $\sum_{\mu} V^{\mu} W^{\mu}$, or a repeated index appearing twice upstairs, is not a slightly informal notation for something correct. It is a *symptom*: either an index has been raised or lowered without saying so, or the expression is wrong.

This is a genuinely useful diagnostic. When you have made an error in a long index computation, the odds are excellent that you can find it without checking any algebra, simply by scanning for an index that is repeated in the same position or that appears three times. Treat the convention as a type system, and let it fail loudly.

▼ Grind box — the quotient theorem, or how to prove something *is* a tensor without transforming it

Often you meet an array and want to know whether it is a tensor, and transforming it directly is painful. There is a shortcut, and it is used constantly.

Claim. Suppose $K_{\mu\nu}$ is an array of sixteen numbers per frame, and suppose that for *every* vector V^{ν} the sixteen-fold sum $K_{\mu\nu} V^{\nu}$ is a covector. Then $K_{\mu\nu}$ is a $(0, 2)$ tensor.

Proof. By hypothesis $W_{\mu} \equiv K_{\mu\nu} V^{\nu}$ is a covector, so

$$K'_{\mu\nu} V'^{\nu} = W'_{\mu} = (\Lambda^{-1})^{\alpha}_{\mu} W_{\alpha} = (\Lambda^{-1})^{\alpha}_{\mu} K_{\alpha\beta} V^{\beta}.$$

On the left substitute $V'^{\nu} = \Lambda^{\nu}_{\beta} V^{\beta}$:

$$\left[K'_{\mu\nu} \Lambda^{\nu}_{\beta} - (\Lambda^{-1})^{\alpha}_{\mu} K_{\alpha\beta} \right] V^{\beta} = 0 \quad \text{for every } V^{\beta}.$$

A bracket that annihilates every vector is zero — take V to be each of the four basis vectors in turn. Contract the resulting identity with $(\Lambda^{-1})^{\beta}_{\sigma}$ to free the index ν , and you get

$$K'_{\mu\sigma} = (\Lambda^{-1})^{\alpha}_{\mu} (\Lambda^{-1})^{\beta}_{\sigma} K_{\alpha\beta},$$

the $(0, 2)$ law. ■

The same argument works for any index structure. **Why it matters:** this is how you certify that the objects physics hands you are tensors. In Chapter 2.6 the field $F^{\mu\nu}$ is identified by the requirement that $f^{\mu} = q F^{\mu\nu} u_{\nu} / c$ be a four-vector for every four-velocity u_{ν} — and the quotient theorem then forces $F^{\mu\nu}$ to be a tensor, which is most of the work of that chapter done in advance.

One caution, and it is the whole content of the phrase "for every": the hypothesis must hold for *all* vectors, not merely for the one you happen to care about. $\eta_{\mu\nu}N^\mu p^\nu$ in §1 was a number for each particular p ; that never made N^μ a vector.

6 · The whole point, in one theorem

Everything so far has been apparatus. Here is what it was for.

THEOREM — TENSOR EQUATIONS ARE FRAME-INDEPENDENT

Let S and T be tensors of the same type (p, q) . If

$$T^{\mu\nu\cdots}_{\rho\sigma\cdots} = S^{\mu\nu\cdots}_{\rho\sigma\cdots}$$

holds in **one** coordinate system, it holds in **every** coordinate system.

Proof. Define $D \equiv T - S$. By operation (1) of §5.1, D is a tensor of type (p, q) . The hypothesis says every component of D vanishes in the unprimed frame. Now apply the transformation law: each primed component of D is a sum of terms, and every term contains a factor $D^{\alpha\beta\cdots}_{\gamma\delta\cdots} = 0$. A sum of terms each containing a factor of zero is zero. Hence $D' = 0$ in the primed frame too, i.e. $T' = S'$. Since the primed frame was arbitrary, the equation holds everywhere. ■

That proof is four lines and it is not deep. The transformation law is *linear and homogeneous* — every term has exactly one factor of D and no constant term — so it maps zero to zero. That is the entire mechanism. What is remarkable is not the proof but the consequence.

6.1 · What this buys

Consider what you would otherwise have to do. You have a candidate law of physics. You want to know whether it is consistent with relativity — whether an observer sailing past at $0.9c$ would write the same law. The honest procedure is: express everything in the moving observer's coordinates, substitute, grind, and see whether the result has the same form. That is a long calculation, it must be redone for every law, and it is easy to get wrong.

The theorem replaces all of that with an *inspection*. If both sides of your equation are tensors of the same type — if the free indices match, up with up and down with down, on both sides — then you are done. No calculation. The frame-independence is visible in the shape of the equation. This property has a name, **manifest covariance**, and "manifest" is the operative word: not merely true, but true in a way you can see.

So the rest of this book has a writing convention that is also a physical principle:

THE RULE THE REST OF THE BOOK OBEYS

Every fundamental law is written as an equation between tensors of the same type. Anything that cannot be written that way is either not fundamental, or not right, or is hiding a choice of frame that will eventually cost you.

6.2 · The scoreboard

$\mathbf{F} = m\mathbf{a}$ is not a tensor equation. Chapter 1.1 §4.2 already showed the damage from a different direction — that Newton's second law changes form under an innocent change of spatial coordinates, sprouting terms that look like forces and are not. Relativity makes the failure sharper. The left side is a three-component object; the right side involves $d^2\mathbf{x}/dt^2$, in which t is a coordinate rather than a scalar, and Chapter 2.2 established that t is not shared between frames. There is no way to read (2.4.9) as acting on this equation, because the equation is not built from objects the matrix knows how to act on. It is not that $\mathbf{F} = m\mathbf{a}$ is *false* — at $10^{-6}c$ it is superb — it is that its content changes when you change frames, which disqualifies it as a statement about nature rather than about a laboratory.

$\frac{dp^\mu}{d\tau} = f^\mu$ is one. Both sides are $(1, 0)$: p^μ is a vector, $d\tau$ is a scalar, so the derivative is a vector, and f^μ is defined to be whatever four-vector sits on the right. Chapter 2.5 builds it and shows that its spatial part reduces to $\mathbf{F}$ at low speed, so nothing is lost — the old law is recovered as a limit, exactly as $\frac{1}{2}mv^2$ was recovered in Chapter 0.1.

$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu$ is one. Both sides are $(1, 0)$ — the left after contraction. That single equation contains two of Maxwell's four equations, and Chapter 2.6's punchline is that writing electromagnetism this way is not a repackaging but an explanation: the theory was relativistic before relativity was invented, which is why it broke Galilean invariance in Chapter 2.1.

$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$ is one. Both sides are $(0, 2)$ and symmetric. Chapter 3.6 derives it. The reason it can be *guessed*, before it is derived, is this chapter: demand a $(0, 2)$ symmetric tensor built from the metric and its first two derivatives, and there are almost no candidates. Tensor structure does not merely check laws; it drastically narrows the field of possible ones. That is the method the whole second half of this book runs on.

△ TWO THINGS THE THEOREM DOES NOT SAY

It does not say tensor equations are true. $p^\mu = 7 u^\mu$, in whatever units you like, is a perfectly good tensor equation, and it is false for every particle whose rest mass is not 7. Covariance is a constraint on the *form* of a law, not evidence for its content. It narrows the search; experiment still decides.

It does not license equations between components. " $T^{00} = 5$ " is not a tensor equation: the left side is one component of a tensor, the right side is a number, and they do not transform alike. Neither is " $V^1 = V^2$ ", nor " $\mathbf{p} = 0$ ". Such statements can be perfectly true — a particle really can be at rest — but they are statements about a frame, and they must be labelled as such and never carried across a boost unexamined. That is precisely the sin of (2.4.1).

7 · Symmetry, antisymmetry, and a promise

Chapter 0.7 §4.3 proved that any square matrix splits uniquely into a symmetric and an antisymmetric part, and then cashed it: applied to the Jacobian of a velocity field, the symmetric part is strain and the antisymmetric part is rotation, at twice the local angular velocity. The same split applies to any rank-2 tensor, verbatim:

$$T^{\mu\nu} = \underbrace{\frac{1}{2}(T^{\mu\nu} + T^{\nu\mu})}_{T^{(\mu\nu)}} + \underbrace{\frac{1}{2}(T^{\mu\nu} - T^{\nu\mu})}_{T^{[\mu\nu]}}, \quad (2.4.27)$$

with round brackets denoting the symmetric part and square brackets the antisymmetric part — standard notation, worth learning now because Chapter 3.4 uses it heavily. Existence and uniqueness are Chapter 0.7's argument unchanged; nothing about it used three dimensions or a metric.

7.1 · The split is frame-independent

Here is what is new, and it is the reason the split is physics rather than bookkeeping. Suppose $T^{\mu\nu} = T^{\nu\mu}$ in one frame. Transform:

$$T'^{\nu\mu} = \Lambda^\nu_\beta \Lambda^\mu_\alpha T^{\beta\alpha} = \Lambda^\mu_\alpha \Lambda^\nu_\beta T^{\alpha\beta} = T'^{\mu\nu}, \quad (2.4.28)$$

The first equality is the $(2, 0)$ law with the two dummy indices named β, α instead of α, β — a free choice, by §8. The second uses only that numbers commute and that $T^{\beta\alpha} = T^{\alpha\beta}$ by hypothesis. Symmetry survives. The identical computation with one minus sign shows antisymmetry survives. So:

WHY THIS MATTERS

" T is antisymmetric" is a **frame-independent property of the object**, not an accident of the coordinates you chose. Every observer agrees. The decomposition (2.4.27) therefore splits a tensor into two pieces that no coordinate change can mix — and pieces that cannot mix are pieces that can carry different physics.

⚠ ONLY SAME-POSITION INDICES MAY BE SWAPPED

The statement " $T^{\mu\nu}$ is symmetric" is meaningful. The statement " $T^\mu{}_\nu$ is symmetric" is not: you would be comparing $T^\mu{}_\nu$ with $T^\nu{}_\mu$, and those two arrays transform differently, so the comparison holds in one frame and fails in another. If you need such a statement, lower the upper index first with η and then swap. Symmetry is a property of a pair of slots of the *same kind*.

7.2 · Counting, and what the counts are hiding

How many independent components does each piece have in n dimensions? A symmetric array is fixed by its diagonal (n entries) plus its strictly upper triangle ($\frac{1}{2}n(n-1)$ entries):

$$\# \text{symmetric} = \frac{n(n+1)}{2}, \quad \# \text{antisymmetric} = \frac{n(n-1)}{2}, \quad (2.4.29)$$

the antisymmetric count being the strictly upper triangle alone, since the diagonal is forced to zero by $A^{\mu\mu} = -A^{\mu\mu}$ for each fixed μ (no sum). The two add to n^2 , as they must.

Now put numbers in, and watch a coincidence turn into a lifetime of confusion.

In three dimensions, $\frac{1}{2} \cdot 3 \cdot 2 = 3$. An antisymmetric 3×3 array has exactly three independent components — the same number as a vector. So it can *masquerade* as a vector, and in three-dimensional physics it invariably does. Chapter 0.7 §4.4 showed you the mechanism: the curl is really the antisymmetric part of a Jacobian, and it is only because $3 = \frac{1}{2} \cdot 3 \cdot 2$ that we can package it as an arrow at all. The magnetic field is the same trick. $\mathbf{B}$ is not a vector in the way $\mathbf{v}$ is; it is an antisymmetric rank-2 object wearing a disguise that fits only in three dimensions — which is why it behaves oddly under reflection, and why the right-hand rule needs a convention where the laws of physics should not.

In four dimensions, $\frac{1}{2} \cdot 4 \cdot 3 = 6$. Six. Not four, so the disguise fails: an antisymmetric rank-2 tensor in spacetime cannot possibly be a four-vector. It is its own kind of thing, with six slots.

Six independent numbers, sitting in the most natural rank-2 object spacetime admits. And the reader who has been kept waiting since Chapter 0.7 §4.4 should now be doing the arithmetic: three components of $\mathbf{E}$, three components of $\mathbf{B}$, total six.

THE PROMISE

Chapter 2.6 will find exactly six components sitting in an antisymmetric $F^{\mu\nu}$, and they will be the three components of $\mathbf{E}$ and the three of $\mathbf{B}$. Not by analogy — by derivation. $\mathbf{E}$ and $\mathbf{B}$ are not two fields that happen to be coupled; they are one tensor, split up differently by different observers, exactly as ρ and $\mathbf{J}$ were one four-vector and t and $\mathbf{x}$ were one set of coordinates.

Nothing more will be said about it here. But you should now find that outcome unsurprising, and finding it unsurprising in advance is the point of this section.

One more count for the road: $\frac{1}{2} \cdot 4 \cdot 5 = 10$. A *symmetric* rank-2 tensor in four dimensions has ten independent components. That is the metric $g_{\mu\nu}$ of Chapter 3.3, and it is why "the gravitational field" turns out to be ten functions rather than one. Problem 4 asks you to do these counts yourself, because doing them is how they stick.

8 · Notation hygiene

Short section, high yield. These rules are not style preferences; each one is a consequence of something already proved, and each violation is a detectable error.

Free indices must match on both sides. A *free* index is one that appears exactly once in a term. It labels which component you are talking about, so both sides must be labelled the same way — same letters, same positions, same

order of appearance. $A^\mu = B^\mu$ is fine. $A^\mu = B_\mu$ is not (different species, §3.3). $A^\mu = B^{\mu\nu}$ is not (the right side is sixteen numbers per value of μ). Every term in a sum must carry the same free indices, too: $A^\mu + B^\nu$ is meaningless.

Dummy indices are summed and freely renameable. A *dummy* index appears exactly twice, once up and once down, and is summed. Its name is invisible to the result: $V^\mu \omega_\mu = V^\alpha \omega_\alpha$. Use that freedom aggressively — in a long computation, renaming dummies to keep them distinct is the difference between an answer and a mess.

An index may not appear three times. If it does, either you meant two different indices and reused a letter, or you have a genuine error. There is no valid expression in which a letter appears three times in one term. This rule catches more mistakes than any other, precisely because it is purely mechanical: you can check it without understanding the expression at all.

The Kronecker delta is the mixed-index identity. $\delta^\mu{}_\nu = 1$ if $\mu = \nu$ and 0 otherwise. It has one index up and one down because it is a $(1, 1)$ tensor — Problem 1 asks you to verify that, and the answer is more interesting than it sounds. Its two working properties are that it renames an index, $\delta^\mu{}_\nu V^\nu = V^\mu$, and that its trace counts dimensions:

$$\delta^\mu{}_\mu = 1 + 1 + 1 + 1 = 4. \quad (2.4.30)$$

Not 1. Getting that wrong is a rite of passage; the number 4 will appear all over Chapter 3.4 and, in the guise of d , throughout Chapter 5.9's dimensional regularisation.

∂_μ **carries a lower index.** That is (2.4.11), and it is the most frequently forgotten fact in the subject, so here it is a second time: $\partial_\mu = \frac{\partial}{\partial x^\mu}$ carries a *lower* index, even though x^μ carries an upper one — because differentiating with respect to a thing inverts how it transforms. If you find yourself writing ∂^μ , you must have raised it with the metric, and there is a minus sign on the spatial components waiting for you.

8.1 · The Levi-Civita symbol, and an honest caveat

One more array is worth naming. Define $\epsilon_{\mu\nu\rho\sigma}$ to be totally antisymmetric — it changes sign under the exchange of any two indices — with $\epsilon_{0123} = +1$. Total antisymmetry means it vanishes unless all four indices are different, so its 256 entries are ± 1 on the 24 permutations of $(0, 1, 2, 3)$ and zero elsewhere.

How does it transform? First a fact about any 4×4 matrix M , taken straight from Chapter 0.4 §5: the determinant is *the* alternating multilinear function of the columns normalised by $\det I = 1$, so contracting a totally antisymmetric array with four copies of M can only reproduce that array times the determinant,

$$M^\alpha{}_\mu M^\beta{}_\nu M^\gamma{}_\rho M^\delta{}_\sigma \epsilon_{\alpha\beta\gamma\delta} = \det(M) \epsilon_{\mu\nu\rho\sigma}. \quad (2.4.31)$$

Now ask what the $(0, 4)$ tensor law would do to ϵ . That law uses four factors of Λ^{-1} , so set $M = \Lambda^{-1}$ in (2.4.31):

$$\epsilon'_{\mu\nu\rho\sigma} = \det(\Lambda^{-1}) \epsilon_{\mu\nu\rho\sigma} = \frac{1}{\det \Lambda} \epsilon_{\mu\nu\rho\sigma}. \quad (2.4.32)$$

By the grind box in §4, $\det \Lambda = \pm 1$ for Lorentz transformations, so $1/\det \Lambda = \det \Lambda$ and the factor is ± 1 . Under boosts and rotations ($\det = +1$) the symbol is genuinely invariant and may be used exactly like a $(0, 4)$ tensor;

under a spatial reflection ($\det = -1$) it flips sign. That is where "pseudovector" and "pseudoscalar" come from, and Chapter 5.4 will find it decisive when the weak interaction turns out to care about the difference.

■ For a *general* coordinate change $\det \Lambda$ is neither ± 1 nor constant, and (2.4.32) then says that an ϵ with fixed entries is not a tensor at all but a *tensor density* — an object that picks up a power of the Jacobian determinant. The repair is to multiply by $\sqrt{-g}$, where $g = \det g_{\mu\nu}$, which carries a compensating determinant. We quote that here and defer it: it is derived in Chapters 3.3 and 3.5, where it is also what makes $\sqrt{-g} d^4x$ the invariant volume element. Within Part II, where $\det \Lambda = \pm 1$, none of this bites.

9 • Worked examples

WORKED EXAMPLE 1 — BOOSTING A FOUR-MOMENTUM TWICE, AND THE NUMBER THAT WILL NOT MOVE

Take $p^\mu = (5, 3, 0, 0)$ in units of MeV/c . Boost it into two different frames and verify $p_\mu p^\mu$ by hand each time.

Frame S . Lower the index by (2.4.23): flip the spatial signs, leaving $p_\mu = (5, -3, 0, 0)$. Contract:

$$p_\mu p^\mu = (5)(5) + (-3)(3) + 0 + 0 = 25 - 9 = 16.$$

So $mc = 4 \text{ MeV}/c$. For contrast, compute the forbidden pairing $\sum_\mu p^\mu p^\mu$, with both indices up: $25 + 9 = 34$. Remember that number.

Frame S' : $\beta = 3/5$, so $\gamma = (1 - 9/25)^{-1/2} = 5/4$. Apply (2.4.9) row by row:

$$p'^0 = \gamma(p^0 - \beta p^1) = \frac{5}{4}(5 - \frac{3}{5} \cdot 3) = \frac{5}{4} \cdot \frac{16}{5} = 4,$$

$$p'^1 = \gamma(p^1 - \beta p^0) = \frac{5}{4}(3 - \frac{3}{5} \cdot 5) = \frac{5}{4} \cdot 0 = 0,$$

with $p'^2 = p'^3 = 0$ untouched. So $p'^\mu = (4, 0, 0, 0)$ — this is the particle's *rest frame*, which is no accident: the particle's velocity in S is $v/c = p^1 c/E = 3/5$, exactly the boost we chose. Contract:

$$p'_\mu p'^\mu = (4)(4) - 0 - 0 - 0 = 16. \checkmark$$

The forbidden pairing here gives $16 + 0 = 16$.

Frame S'' : boost S' again, by $\beta_2 = -4/5$, so $\gamma_2 = 5/3$.

$$p''^0 = \frac{5}{3}(4 - (-\frac{4}{5}) \cdot 0) = \frac{20}{3},$$

$$p''^1 = \frac{5}{3}(0 - (-\frac{4}{5}) \cdot 4) = \frac{5}{3} \cdot \frac{16}{5} = \frac{16}{3}.$$

Contract, and note that the two large numbers must conspire:

$$p''_\mu p''^\mu = \left(\frac{20}{3}\right)^2 - \left(\frac{16}{3}\right)^2 = \frac{400 - 256}{9} = \frac{144}{9} = 16. \checkmark$$

The forbidden pairing gives $\frac{400+256}{9} = \frac{656}{9} = 72.888\dots$

The scoreboard.

FRAME	p^μ	$p_\mu p^\mu$	$\sum_\mu p^\mu p^\mu$ (wrong)
S	$(5, 3, 0, 0)$	16	34
S'	$(4, 0, 0, 0)$	16	16
S''	$(\frac{20}{3}, \frac{16}{3}, 0, 0)$	16	$\frac{656}{9}$

The correctly contracted column is constant and the naively contracted one is not. That column is (2.4.13) and this chapter, in a table.

A check on the framework. Rapidities add, by the grind box in §2. Here $\phi_1 = \text{artanh } \frac{3}{5} = \frac{1}{2} \ln \frac{1+3/5}{1-3/5} = \frac{1}{2} \ln 4 = \ln 2$ and $\phi_2 = \text{artanh}(-\frac{4}{5}) = \frac{1}{2} \ln \frac{1}{9} = -\ln 3$, so the composite rapidity is $\ln 2 - \ln 3 = \ln \frac{2}{3}$ and the composite velocity is

$$\beta_{\text{tot}} = \tanh \left(\ln \frac{2}{3} \right) = \frac{(2/3)^2 - 1}{(2/3)^2 + 1} = \frac{-5/9}{13/9} = -\frac{5}{13}, \quad \gamma_{\text{tot}} = \frac{13}{12}.$$

Boost the *original* $p^\mu = (5, 3, 0, 0)$ by that single transformation:

$$\begin{aligned} \frac{13}{12} \left(5 + \frac{5}{13} \cdot 3 \right) &= \frac{13}{12} \cdot \frac{80}{13} = \frac{20}{3}, \\ \frac{13}{12} \left(3 + \frac{5}{13} \cdot 5 \right) &= \frac{13}{12} \cdot \frac{64}{13} = \frac{16}{3}, \end{aligned}$$

which is p''^μ exactly. Two boosts composed equal one boost at the added rapidity, and the invariant is 16 throughout — as it must be, since it never depended on Λ at all.

WORKED EXAMPLE 2 — $\partial_M V^M$ IS A SCALAR, AND WHAT THAT DOES TO CHARGE CONSERVATION

Show that $\partial_\nu V^\mu$ is a $(1, 1)$ tensor and hence that $\partial_\mu V^\mu$ is a scalar. Then read Chapter 0.7's continuity equation in that light.

Step 1. Differentiate the vector law. Using the covector law (2.4.11) for ∂'_ν and then the vector law for V'^μ :

$$\partial'_\nu V'^\mu = (\Lambda^{-1})^\beta{}_\nu \partial_\beta (\Lambda^\mu{}_\alpha V^\alpha).$$

Now use the fact recorded under (2.4.9): for a Lorentz transformation every entry of Λ is a constant, so ∂_β passes straight through it:

$$\partial'_\nu V'^\mu = \Lambda^\mu{}_\alpha (\Lambda^{-1})^\beta{}_\nu \partial_\beta V^\alpha,$$

which is exactly the $(1, 1)$ transformation law. So $\partial_\nu V^\mu$ is a tensor: sixteen numbers that transform correctly.

Step 2. Contract μ with ν . By §5.1's theorem, contracting a $(1, 1)$ gives a $(0, 0)$ — a scalar. So

$$\partial_\mu V^\mu \text{ is the same number in every inertial frame.}$$

The four-divergence is an invariant. Note what is *not* claimed: $\partial_0 V^0$ alone is not invariant, and neither is $\partial_i V^i$. Only the sum, with the index correctly paired, is.

Step 3 — cashing it. Chapter 0.7 §6 derived the continuity equation $\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0$ from nothing but "stuff is neither created nor destroyed", and noted at the end that ρ and $\mathbf{J}$ assemble into $J^\mu = (\rho c, \mathbf{J})$. Take that seriously and expand $\partial_\mu J^\mu$, remembering $x^0 = ct$ so $\partial_0 = \frac{\partial}{\partial x^0} = \frac{1}{c} \frac{\partial}{\partial t}$:

$$\partial_\mu J^\mu = \partial_0 J^0 + \partial_i J^i = \frac{1}{c} \frac{\partial}{\partial t} (\rho c) + \nabla \cdot \mathbf{J} = \frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J}.$$

The factors of c cancel exactly, which is the arithmetic reason J^μ is defined with the c in the time slot. So

$$\boxed{\partial_\mu J^\mu = 0}$$

is *the same equation* as Chapter 0.7's continuity equation. Not an analogue: the same equation, rewritten.

But it now says more than it did. In Chapter 0.7 it was an equation about a particular observer's ρ and $\mathbf{J}$, in a particular frame, with a time derivative singled out. Here the left side is a scalar and the right side is zero — a scalar equation — so by §6 it holds in every frame with the same form. **If charge is conserved for one observer it is conserved for all of them**, and the two-term split into "rate of change here" and "flow away from here" is just how each observer happens to slice a single four-dimensional statement. Chapter 0.7 could not have told you that; it did not have the machinery. The whole content of the upgrade is the index placement.

▼ Grind box — where Step 1 breaks, and why Part III needs a new derivative

Step 1 above used one thing beyond the definitions: that $\Lambda^\mu{}_\alpha$ is constant, so ∂_β passes through it. Suppose it is not — as it is not for a change to polar coordinates, or an accelerating frame, or any coordinate patch on a curved manifold. Then the product rule gives an extra term:

$$\partial'_\nu V'^\mu = (\Lambda^{-1})^\beta{}_\nu \left[\Lambda^\mu{}_\alpha \partial_\beta V^\alpha + \underbrace{(\partial_\beta \Lambda^\mu{}_\alpha) V^\alpha}_{\text{inhomogeneous}} \right].$$

The first term is the tensor law. The second is not proportional to ∂V at all — it is proportional to V itself — so the whole expression is *not* of the form (matrix) $\times$ (components of ∂V). $\partial'_\nu V'^\mu$ **is not a tensor under general coordinate changes**.

Look closely at why this is fatal rather than annoying. The failure is an *inhomogeneous* term, and §6's theorem needed the transformation law to be homogeneous. With an inhomogeneous term, an expression that vanishes in one frame need not vanish in another — the proof of the central theorem collapses, and with it the guarantee that makes tensor equations worth writing.

The repair is to invent a new derivative ∇_μ which differs from ∂_μ by a correction term chosen so that the two inhomogeneous pieces cancel. The correction is called the **connection**, it is built from derivatives of the metric, and constructing it is Chapter 3.3. Its failure to commute with itself is curvature, which is Chapter 3.4, which is gravity.

So: the flat-space simplification that let this chapter be short is exactly the thing Part III gives up, and everything hard about general relativity is downstream of one product rule.

10 · Your turn

PROBLEM 1 — Δ VERSUS H

(a) Show that δ^μ_ν — defined to be 1 when $\mu = \nu$ and 0 otherwise, in every coordinate system — really is a $(1, 1)$ tensor, for *any* invertible coordinate change, not just Lorentz ones. Note that this is the same style of definition that failed disastrously for N^μ in (2.4.1); explain why it succeeds here.

(b) $\eta_{\mu\nu}$ also has the same components in every *inertial* frame. Show this is a special property of Lorentz transformations and not a general one, by exhibiting a coordinate change under which the components of η change.

– Solution

(a) Apply the $(1, 1)$ law to the array δ and see what comes out:

$$\Lambda^\mu_\alpha (\Lambda^{-1})^\beta_\nu \delta^\alpha_\beta = \Lambda^\mu_\alpha (\Lambda^{-1})^\alpha_\nu = \delta^\mu_\nu,$$

using (2.4.12) in the last step. The transformation law maps the array δ to itself, so declaring it to be δ in every frame is *consistent* with the tensor law. And no step used any property of Λ beyond invertibility, so this holds for every smooth coordinate change whatsoever.

Why this succeeds where (2.4.1) failed. The two definitions have the same form — “these components, in every frame” — but only one of them is compatible with the transformation law. Declaring components in all frames is not automatically wrong; it is *an extra condition*, and it is legitimate exactly when the transformation law happens to reproduce those components. For δ it does. For $(1, 0, 0, 0)$ it does not: boosting it gives $(\gamma, -\gamma\beta, 0, 0) \neq (1, 0, 0, 0)$. The test is mechanical, and you should run it whenever you are tempted to fix components in all frames.

(b) By the same test: (2.4.16), $\eta_{\mu\nu} \Lambda^\mu_\rho \Lambda^\nu_\sigma = \eta_{\rho\sigma}$, says exactly that the $(0, 2)$ law maps η to itself. But that identity *is the definition of a Lorentz transformation*. It is not available for other coordinate changes.

Counterexample: rescale one spatial axis, $x'^1 = 2x^1$, everything else unchanged. Then $\Lambda^1_1 = 2$ and $(\Lambda^{-1})^1_1 = \frac{1}{2}$, so

$$\eta'_{11} = (\Lambda^{-1})^\alpha_1 (\Lambda^{-1})^\beta_1 \eta_{\alpha\beta} = \left(\frac{1}{2}\right)^2 \eta_{11} = -\frac{1}{4} \neq -1.$$

The components changed, as they must — measuring x in half-metres has to show up somewhere, and it shows up in the metric. A second, more consequential example: switch the spatial part to plane polar coordinates, $x = r \cos \theta$, $y = r \sin \theta$. Chapter 0.6 §4's grind box computed $dx^2 + dy^2 = dr^2 + r^2 d\theta^2$, so

$$\eta'_{\mu\nu} = \text{diag}(1, -1, -r^2, -1)$$

in coordinates (ct, r, θ, z) — *position-dependent*, in flat spacetime, with no gravity anywhere. That is the punchline worth carrying into Part III: a position-dependent metric does not by itself mean curvature. Telling the two apart requires an object that is zero for the polar metric and nonzero for a genuinely curved one, and constructing it is Chapter 3.4.

PROBLEM 2 — TRANSFORM, THEN CHECK

In frame S , two four-vectors have components $A^\mu = (3, 1, 2, 0)$ and $B^\mu = (2, 1, 0, 1)$. Compute $A \cdot B \equiv \eta_{\mu\nu} A^\mu B^\nu$. Then boost to S' with $\beta = 3/5$, compute all eight primed components, and verify the invariance. Finally compute the naive sum $\sum_\mu A^\mu B^\mu$ in both frames and comment.

– Solution

In S . Lower B : $B_\mu = (2, -1, 0, -1)$. Then

$$A \cdot B = A^\mu B_\mu = (3)(2) + (1)(-1) + (2)(0) + (0)(-1) = 6 - 1 = 5.$$

Boost, $\gamma = 5/4$, $\beta = 3/5$. Only the 0 and 1 components mix:

$$\begin{aligned} A'^0 &= \frac{5}{4} \left(3 - \frac{3}{5} \cdot 1 \right) = \frac{5}{4} \cdot \frac{12}{5} = 3, & A'^1 &= \frac{5}{4} \left(1 - \frac{3}{5} \cdot 3 \right) = \frac{5}{4} \cdot \left(-\frac{4}{5} \right) = -1, \\ B'^0 &= \frac{5}{4} \left(2 - \frac{3}{5} \cdot 1 \right) = \frac{5}{4} \cdot \frac{7}{5} = \frac{7}{4}, & B'^1 &= \frac{5}{4} \left(1 - \frac{3}{5} \cdot 2 \right) = \frac{5}{4} \cdot \left(-\frac{1}{5} \right) = -\frac{1}{4}, \end{aligned}$$

with $A'^2 = 2$, $A'^3 = 0$, $B'^2 = 0$, $B'^3 = 1$ unchanged. So $A'^\mu = (3, -1, 2, 0)$ and $B'^\mu = (\frac{7}{4}, -\frac{1}{4}, 0, 1)$. Contract:

$$A' \cdot B' = (3)\left(\frac{7}{4}\right) - (-1)\left(-\frac{1}{4}\right) - (2)(0) - (0)(1) = \frac{21}{4} - \frac{1}{4} = 5. \checkmark$$

The naive sum. In S : $6 + 1 + 0 + 0 = 7$. In S' : $\frac{21}{4} + \frac{1}{4} + 0 + 0 = \frac{22}{4} = 5.5$. Different. The naive sum is not a scalar, it is not the "length" of anything, and the fact that it is what you would compute in Euclidean space is exactly why the metric has to be written explicitly.

A useful habit: notice that A'^μ and B'^μ are individually quite different from A^μ and B^μ — A^1 even changed sign — while the contraction did not budge. That is the shape of every calculation in relativity worth doing.

PROBLEM 3 — SYMMETRIC AGAINST ANTISYMMETRIC

Let $S^{\mu\nu} = S^{\nu\mu}$ and $A_{\mu\nu} = -A_{\nu\mu}$. Prove that $S^{\mu\nu} A_{\mu\nu} = 0$ identically — no conditions, no special frames. Then say where you expect this to be used.

– Solution

Call the sum $X \equiv S^{\mu\nu} A_{\mu\nu}$. Both indices are dummies, so rename them, swapping the letters: $X = S^{\nu\mu} A_{\nu\mu}$. That is not a manipulation of the object, only of the notation — the same 256 terms in a different order. Now use the two hypotheses on the renamed expression: $S^{\nu\mu} = S^{\mu\nu}$ and $A_{\nu\mu} = -A_{\mu\nu}$, so

$$X = S^{\nu\mu} A_{\nu\mu} = S^{\mu\nu} (-A_{\mu\nu}) = -X.$$

Hence $2X = 0$ and $X = 0$. ■ Note that the proof is pure index bookkeeping: no metric, no frame, no dimension. It holds in any number of dimensions and for any pair of contracted slots.

Where it gets used. Constantly, and usually to make an unwanted term disappear.

- Chapter 2.6: $F^{\mu\nu}$ is antisymmetric, so contracting it with anything symmetric kills the term. In particular $\partial_\mu \partial_\nu F^{\mu\nu} = 0$, because $\partial_\mu \partial_\nu$ is symmetric in $\mu\nu$ by Clairaut's theorem (Chapter 0.6 §6). Applying ∂_ν to $\partial_\mu F^{\mu\nu} = \mu_0 J^\nu$ then forces $\partial_\nu J^\nu = 0$: *Maxwell's equations require charge conservation*, they do not merely permit it. That is this one-line lemma doing real work.
- Chapter 3.3: the connection is symmetric in its lower indices, so contracting it with an antisymmetric object drops out — which is why the exterior derivative of Chapter 3.5 needs no connection at all.
- Chapter 3.6: the Einstein tensor is symmetric, so only the symmetric part of any candidate source can appear — which is a constraint on what the right-hand side of the field equations is allowed to be.

PROBLEM 4 — COUNTING

In n dimensions, count the independent components of a symmetric rank-2 tensor and of an antisymmetric one, and verify the two counts sum to n^2 . Specialise to $n = 4$ and identify what each of the two numbers will turn out to be. Then do $n = 3$ and explain the historical accident that makes the magnetic field look like a vector.

– Solution

Symmetric. $T^{\mu\nu} = T^{\nu\mu}$, so the array is determined by the entries with $\mu \leq \nu$. There are n diagonal entries and $\frac{1}{2}n(n-1)$ strictly-upper ones (half of the $n^2 - n$ off-diagonal entries), giving

$$n + \frac{n(n-1)}{2} = \frac{2n + n^2 - n}{2} = \frac{n(n+1)}{2}.$$

Antisymmetric. $A^{\mu\nu} = -A^{\nu\mu}$ forces $A^{\mu\mu} = -A^{\mu\mu}$ for each fixed μ (no sum), hence $A^{\mu\mu} = 0$: the diagonal is empty. Only the strictly-upper triangle survives: $\frac{1}{2}n(n-1)$.

Sum. $\frac{1}{2}n(n+1) + \frac{1}{2}n(n-1) = \frac{1}{2}n[(n+1) + (n-1)] = n^2$. ✓ — as required, since (2.4.27) writes every rank-2 array as symmetric plus antisymmetric, uniquely, so the two pieces must account for all n^2 numbers exactly once.

$n = 4$. Symmetric: $\frac{1}{2} \cdot 4 \cdot 5 = 10$. Antisymmetric: $\frac{1}{2} \cdot 4 \cdot 3 = 6$. And $10 + 6 = 16 = 4^2$. ✓

The ten is the metric $g_{\mu\nu}$ of Chapter 3.3 — the gravitational field is ten functions, which is why general relativity is ten coupled equations and not one. (Four of the ten are pure coordinate freedom, leaving six genuine ones; that count is Chapter 3.6's business.) The six is the electromagnetic field tensor $F^{\mu\nu}$ of Chapter 2.6: three components of $\mathbf{E}$ and three of $\mathbf{B}$, and now you know in advance that there is exactly enough room and not one slot to spare.

$n = 3$. Symmetric: 6. Antisymmetric: 3. The antisymmetric count coincides with the number of components of a vector — solve $\frac{1}{2}n(n-1) = n$ and you get $n = 3$ (or the trivial $n = 0$), so this happens in three dimensions and nowhere else. That coincidence is why the curl of a vector field can be packaged as a vector (Chapter 0.7 §4.4) and why $\mathbf{B}$ is written as one.

The disguise is detectable, though. An antisymmetric object dressed as a vector reveals itself under reflection: reflect space and a true vector flips its components, while these — built from a determinant-like construction, hence carrying an extra factor of $\det \Lambda = -1$ as in (2.4.32) — do not. Hence "axial vector", "pseudovector", and the right-hand rule, which is a convention we are forced to state because we have insisted on describing a six-slot object with three-slot notation. In four dimensions the problem simply does not arise, because there $6 \neq 4$ and nobody is tempted.

THE BRICK YOU JUST LAID

You now have the definition the rest of the book runs on. **A tensor of type (p, q) is an object whose components transform with p Jacobians and q inverse Jacobians** — not a matrix, not an array, not a picture. Upper and lower indices are two different species: vectors transform like dx^μ , covectors like $\partial_\mu f$, and they pair to give a number that every observer agrees on because the two Jacobians collapse to a δ . You saw that measured, not asserted, in the figure: components moving in opposite directions while their contraction sat at 3.50000000 and the arrow went on piercing exactly three lines.

You have the metric as the dictionary between the species — $V_\mu = \eta_{\mu\nu} V^\nu$, which flips three signs and is a linear map rather than a typographical convenience — and you know that η having fixed components is a defining property of Lorentz transformations, not a law of nature. You have the three operations that stay inside the class, with contraction proved. And you have the theorem the chapter exists for: **a tensor equation true in one frame is true in all of them**, because the transformation law is linear and homogeneous and therefore maps zero to zero. That is why every fundamental law from here on will be written with matched indices, and why $\mathbf{F} = m\mathbf{a}$ cannot be.

Where this gets spent. Immediately, in Chapter 2.5, which writes dynamics as $dp^\mu/d\tau = f^\mu$ and gets $E = \gamma mc^2$ out of a contraction. Then Chapter 2.6, where §7's count of six becomes $F^{\mu\nu}$ and $\mathbf{E}$ and $\mathbf{B}$ stop being two fields. Then Part III, where the definition is reused verbatim with one change — Λ acquires position dependence — and that single change costs you the ordinary derivative (Chapter 3.3's connection), buys you curvature (Chapter 3.4), and produces $G_{\mu\nu} = 8\pi G T_{\mu\nu}/c^4$ (Chapter 3.6), which is a tensor equation and could not have been anything else. Chapter 5.2 writes field theory in this language because it must, and Chapter 6.4 writes Yang–Mills in it because the gauge field is a connection in the same sense as Chapter 3.3's.

Chapter 2.5 now takes the definition and does physics with it.